

MoSPI Reference Manual: Data Analytics, ML & Spatial GIS

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1. Machine Learning for Official Industrial Statistics

Machine learning models (Random Forest, XGBoost) are deployed for automated outlier detection and imputation in Annual Survey of Industries (ASI) microdata. Automated validation pipelines reduce manual inspection overhead while preserving cross-sectoral output consistency.

2. Spatial Data Integration & GIS Mapping

Integration of QGIS and ArcGIS frameworks enables geo-referencing of primary sampling units (PSUs) and household cluster mapping. Spatial interpolation methods (Kriging, IDW) project district-level socio-economic indicators onto continuous spatial grids.

3. Big Data Engineering & API Frameworks

RESTful API architectures facilitate real-time, secure data exchanges between MoSPI databases and external ministry portals. Automated ETL pipelines clean, validate, and convert raw transactional records into standardized JSON payloads for analysis.

4. Open Source Analytics Stack in Official Computing

The official statistical stack prioritizes open-source tools including Python (Pandas, Scikit-Learn), R (tidyverse), PostgreSQL/PostGIS, and Apache Spark for scalable distributed processing of census-scale microdata.